

Module-V

DIFFERENCE EQUATIONS

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Definition

A difference equation is a relation between the independent variable n and dependent variable y and its finite differences.

For example, $y_{n+2} + 6y_{n+1} + 9y_n = 2^n$

Formation of Difference Equation:

Illustrative Examples:

Example 1: Form difference equation by eliminating constants from $U_n = \alpha 2^{n+1}$.

Solution:

$$\text{Given, } U_n = \alpha 2^{n+1} \text{ --- (1)}$$

$$U_{n+1} = \alpha 2^{n+2}$$

$$U_{n+1} = \alpha 2^{n+1} \cdot 2$$

$$U_{n+1} = U_n \cdot 2 \text{ Using (1)}$$

$$U_{n+1} = 2U_n$$

Example 2: Form the difference equation by eliminating the constants A & B from

$$y_n = A \cdot 2^n + B(-2)^n$$

Solution:

$$y_n = A \cdot 2^n + B(-2)^n \text{ --- (1)}$$

$$y_{n+1} = A \cdot 2^{n+1} + B(-2)^{n+1} \text{ --- (2)}$$

$$y_{n+2} = A \cdot 2^{n+2} + B(-2)^{n+2} \text{ --- (3)}$$

Eliminating A and B from (1), (2) and (3) we get,

$$\begin{vmatrix} 2^n & (-2)^n & y_n \\ 2^{n+1} & (-2)^{n+1} & y_{n+1} \\ 2^{n+2} & (-2)^{n+2} & y_{n+2} \end{vmatrix} = 0$$

$$\begin{vmatrix} 2^n & (-2)^n & y_n \\ 2 \cdot 2^n & (-2)(-2)^n & y_{n+1} \\ 2^2 \cdot 2^n & (-2)^2(-2)^n & y_{n+2} \end{vmatrix} = 0$$

$$2^n(-2)^n \begin{vmatrix} 1 & 1 & y_n \\ 2 & -2 & y_{n+1} \\ 4 & 4 & y_{n+2} \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 & 1 & y_n \\ 2 & -2 & y_{n+1} \\ 4 & 4 & y_{n+2} \end{vmatrix} = 0$$

$$1\{-2y_{n+2} - 4y_{n+1}\} - 1\{2y_{n+2} - 4y_{n+1}\} + y_n\{8 + 8\} = 0$$

$$-2y_{n+2} - 4y_{n+1} - 2y_{n+2} + 4y_{n+1} + 16y_n = 0$$

$$-4y_{n+2} + 16y_n = 0$$

$$y_{n+2} - 4y_n = 0$$

Example 3: Find the difference equation from $y_n = (A + Bn)2^n$

Solution: Given, $y_n = (A + Bn)2^n$

$$y_n = A2^n + Bn2^n \text{ --- (1)}$$

$$y_{n+1} = A2^{n+1} + B(n+1)2^{n+1} \text{ --- (2)}$$

$$y_{n+2} = A2^{n+2} + B(n+2)2^{n+2} \text{ --- (3)}$$

Eliminating A and B from (1), (2) and (3) we get

$$\begin{vmatrix} 2^n & n2^n & y_n \\ 2^{n+1} & (n+1)2^{n+1} & y_{n+1} \\ 2^{n+2} & (n+2)2^{n+2} & y_{n+2} \end{vmatrix} = 0$$

$$\begin{vmatrix} 2^n & n2^n & y_n \\ 2 \cdot 2^n & (n+1)2 \cdot 2^n & y_{n+1} \\ 2^2 \cdot 2^n & (n+2)2^2 \cdot 2^n & y_{n+2} \end{vmatrix} = 0$$

$$(2^n)(2^n) \begin{vmatrix} 1 & n & y_n \\ 2 & 2(n+1) & y_{n+1} \\ 4 & 4(n+2) & y_{n+2} \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 & n & y_n \\ 2 & 2(n+1) & y_{n+1} \\ 4 & 4(n+2) & y_{n+2} \end{vmatrix} = 0$$

$$\{2(n+1)y_{n+2} - 4(n+2)y_{n+1}\} - n\{2y_{n+2} - 4y_{n+1}\} + y_n\{8(n+2) - 8(n+1)\} = 0$$

$$2(n+1)y_{n+2} - 4(n+2)y_{n+1} - 2ny_{n+2} + 4ny_{n+1} + y_n\{8n + 16 - 8n - 8\} = 0$$

$$\{2(n+1) - 2n\}y_{n+2} + \{-4(n+2) + 4n\}y_{n+1} + 8y_n = 0$$

$$\{2n + 2 - 2n\}y_{n+2} + \{-4n - 8 + 4n\}y_{n+1} + 8y_n = 0$$

$$2y_{n+2} - 8y_{n+1} + 8y_n = 0$$

$$y_{n+2} - 4y_{n+1} + 4y_n = 0$$

Problems for Practise

1. Form the difference equation from $y_n = A \cdot 5^n$ Ans: $y_{n+1} - 5y_n = 0$
2. Form the difference equation from $y_n = a + b3^n$ Ans: $y_{n+2} - 4y_{n+1} + 3y_n = 0$
3. Form difference equation from $y_n = A \cdot 3^n + B \cdot 5^n$ Ans: $y_{n+2} - 8y_{n+1} + 15y_n = 0$

Application of Z-transform to Solve Linear Difference Equations:

Working Procedure

1. $Z\{y_n\} = F(z)$
2. Apply Z-transform on both sides of the difference equation using the formulae
 $Z\{y_{n+1}\} = z\{F(z) - y_0\}$
 $Z\{y_{n+2}\} = z^2\left\{F(z) - y_0 - \frac{y_1}{z}\right\}$
 $Z\{y_{n+3}\} = z^3\left\{F(z) - y_0 - \frac{y_1}{z} - \frac{y_2}{z^2}\right\}$
3. Use the given conditions and simply to get $F(z)$
4. Express $F(z)$ in terms of Z-transforms of known functions.
5. Take inverse Z-transform on both sides.
6. We use partial fraction method and Residue method to find inverse Z-transforms.
7. This gives y_n as a function of n , which is the desired solution.

Note:

$y_n = y(n)$	$y_0 = y(0)$
$y_{n+1} = y(n+1)$	$y_1 = y(1)$
$y_{n+2} = y(n+2)$	$y_2 = y(2)$
$y_{n+3} = y(n+3), \text{ etc.}$	$y_3 = y(3), \text{ etc.}$

Illustrative Examples:

Example 1: Solve $y_{n+2} + 6y_{n+1} + 9y_n = 2^n$ with $y_0 = y_1 = 0$ using Z-transform

Solution:

Given, $y_{n+2} + 6y_{n+1} + 9y_n = 2^n$

Taking Z-transform on both sides

$$Z\{y_{n+2}\} + 6Z\{y_{n+1}\} + 9Z\{y_n\} = Z\{2^n\}$$

$$z^2\left\{F(z) - y_0 - \frac{y_1}{z}\right\} + 6z\{F(z) - y_0\} + 9F(z) = \frac{z}{z-2} \quad \text{Since, } Z\{a^n\} = \frac{z}{z-a}$$

$$z^2F(z) - z^2y_0 - zy_1 + 6zF(z) - 6zy_0 + 9F(z) = \frac{z}{z-2}$$

Given, $y_0 = y_1 = 0$

$$z^2F(z) - 0 - 0 + 6zF(z) - 0 + 9F(z) = \frac{z}{z-2}$$

$$z^2F(z) + 6zF(z) + 9F(z) = \frac{z}{z-2}$$

$$\{z^2 + 6z + 9\}F(z) = \frac{z}{z-2}$$

$$F(z) = \frac{z}{(z-2)(z^2 + 6z + 9)}$$

$$F(z) = Z\{y_n\} = \frac{z}{(z-2)(z+3)^2}$$

To find y_n using Residue Method

$$\text{Let } F(z)z^{n-1} = z^{n-1} \frac{z}{(z-2)(z+3)^2}$$

$$F(z)z^{n-1} = \frac{z^n}{(z-2)(z+3)^2}$$

The poles of $F(z)z^{n-1}$ are given by $(z-2)(z+3)^n = 0$

Therefore, $z = 2$ is Simple pole

$z = -3$ is pole of order 2.

$$R_1 = \{Res F(z)z^{n-1}\}_{z=a} = \lim_{z \rightarrow a} (z-a) F(z)z^{n-1}$$

Here, $a = 2$

$$R_1 = \lim_{z \rightarrow 2} (z-2) \frac{z^n}{(z-2)(z+3)^2}$$

$$= \lim_{z \rightarrow 2} \frac{z^n}{(z+3)^2} = \frac{2^n}{(2+3)^2}$$

$$R_1 = \frac{1}{25} 2^n$$

$$R_2 = \{Res F(z)z^{n-1}\}_{z=a} = \frac{1}{1!} \lim_{z \rightarrow a} \frac{d}{dz} \{(z-a)^2 F(z)z^{n-1}\}$$

Here, $a = -3$

$$R_2 = \frac{1}{1!} \lim_{z \rightarrow -3} \frac{d}{dz} \left[(z+3)^2 \frac{z^n}{(z-2)(z+3)^2} \right]$$

$$= \lim_{z \rightarrow -3} \frac{d}{dz} \left[\frac{z^n}{(z-2)} \right]$$

$$= \lim_{z \rightarrow -3} \left[\frac{(z-2)nz^{n-1} - z^n(1)}{(z-2)^2} \right]$$

$$= \frac{(-3-2)n(-3)^{n-1} - (-3)^n}{(-3-2)^2} = \frac{(-5)n(-3)^{n-1} - (-3)^n}{(-5)^2}$$

$$= \frac{1}{25} \{-5n(-3)^{n-1} - (-3)^n\}$$

$$R_2 = \frac{-1}{25} \{5n(-3)^{n-1} + (-3)^n\}$$

By method of residues, $y_n = \text{Sum of Residues}$

$$= R_1 + R_2$$

$$y_n = \frac{1}{25} 2^n - \frac{1}{25} \{5n(-3)^{n-1} + (-3)^n\}$$

Example 2: Solve the difference equation $y_{n+3} - 3y_{n+1} + 2y_n = 0$ given that $y(0) =$

4, $y(1) = 0$ and $y(2) = 8$

Solution: Given, $y_{n+3} - 3y_{n+1} + 2y_n = 0$

Taking Z-transform on both sides

$$Z\{y_{n+3}\} - 3Z\{y_{n+1}\} + 2Z\{y_n\} = Z\{0\}$$

$$z^3 \left\{ F(z) - y_0 - \frac{y_1}{z} - \frac{y_2}{z^2} \right\} - 3z\{F(z) - y_0\} + 2F(z) = 0$$

$$z^3 F(z) - z^3 y_0 - z^2 y_1 - z y_2 - 3z F(z) + 3z y_0 + 2F(z) = 0$$

Given that, $y(0) = y_0 = 4$; $y(1) = y_1 = 0$; $y(2) = y_2 = 8$

$$z^3 F(z) - 4z^3 - 0 - 8z - 3z F(z) + 12z + 2F(z) = 0$$

$$z^3 F(z) - 3z F(z) + 2F(z) - 4z^3 + 4z = 0$$

$$\{z^3 - 3z + 2\} F(z) = 4z^3 - 4z$$

$$F(z) = \frac{4z(z^2 - 1)}{z^3 - 3z + 2}$$

$$F(z) = Z\{y_n\} = \frac{4z(z-1)(z+1)}{(z-1)^2(z+2)}$$

$$F(z) = Z\{y_n\} = \frac{4z(z+1)}{(z-1)(z+2)}$$

To find y_n using Residue Method

$$\text{Let } F(z)z^{n-1} = z^{n-1} \frac{4z(z+1)}{(z-1)(z+2)}$$

$$F(z)z^{n-1} = \frac{z^n 4(z+1)}{(z-1)(z+2)}$$

The poles of $F(z)z^{n-1}$ are given by $(z-1)(z+2) = 0$

Therefore, $z = 1, -2$ are simple poles.

$$R_1 = \{Res F(z)z^{n-1}\}_{z=a} = \lim_{z \rightarrow a} (z-a) F(z)z^{n-1}$$

Here, $a = 1$

$$R_1 = \lim_{z \rightarrow 1} (z-1) \frac{z^n 4(z+1)}{(z-1)(z+2)}$$

$$= \lim_{z \rightarrow 1} \frac{z^n 4(z+1)}{(z+2)}$$

$$z^3 - 3z + 2 = z^3 + 0z^2 - 3z + 2$$

	1	0	-3	2
1	0	1	1	-2
	1	1	-2	0

$$z^3 - 3z + 2 = (z-1)(z^2 + z - 2)$$

$$= (z-1)(z^2 + z - 2)$$

$$= (z-1)(z-1)(z+2)$$

$$z^3 - 3z + 2 = (z-1)^2(z+2)$$

$$= \frac{(1)^n 4(1+1)}{(1+2)} = \frac{8}{3} (1)^n$$

$$R_1 = \frac{8}{3}$$

Here, $a = -2$

$$\begin{aligned} R_2 &= \lim_{z \rightarrow -2} (z+2) \frac{z^n 4(z+1)}{(z-1)(z+2)} \\ &= \lim_{z \rightarrow -2} \frac{z^n 4(z+1)}{(z-1)} \\ &= \frac{(-2)^n 4(-2+1)}{(-2-1)} = \frac{4(-1)}{(-3)} (-2)^n \\ R_2 &= \frac{4}{3} (-2)^n \end{aligned}$$

By method of residues, $y_n = \text{Sum of Residues}$

$$= R_1 + R_2$$

$$y_n = y(n) = \frac{8}{3} + \frac{4}{3} (-2)^n$$

Example 3: Solve the difference equation $y_{n+2} + 2y_{n+1} + y_n = n$ given that $y_0 =$

$$y_1 = 0$$

Solution:

$$\text{Given, } y_{n+2} + 2y_{n+1} + y_n = n$$

Taking Z-transform on both sides

$$Z\{y_{n+2}\} + 2Z\{y_{n+1}\} + Z\{y_n\} = Z\{n\}$$

$$z^2 \left\{ F(z) - y_0 - \frac{y_1}{z} \right\} + 2z \{ F(z) - y_0 \} + F(z) = Z\{n\} \quad \text{Since, } Z\{n\} = \frac{z}{(z-1)^2}$$

$$z^2 F(z) - z^2 y_0 - z y_1 + 2z F(z) - 2z y_0 + F(z) = \frac{z}{(z-1)^2}$$

Given that, $y_0 = y_1 = 0$

$$z^2 F(z) - 0 - 0 + 2z F(z) - 0 + F(z) = \frac{z}{(z-1)^2}$$

$$\{z^2 + 2z + 1\} F(z) = \frac{z}{(z-1)^2}$$

$$F(z) = \frac{z}{(z-1)^2 \{z^2 + 2z + 1\}}$$

$$F(z) = Z\{y_n\} = \frac{z}{(z-1)^2 (z+1)^2}$$

To find y_n using Residue Method

$$\text{Let } F(z)z^{n-1} = z^{n-1} \left\{ \frac{z}{(z-1)^2 (z+1)^2} \right\}$$

$$F(z)z^{n-1} = \frac{z^n}{(z-1)^2 (z+1)^2}$$

The poles of $F(z)z^{n-1}$ are given by $(z-1)^2 (z+1)^2 = 0$

Therefore, $z = 1$ is pole of order 2.

$z = -1$ is pole of order 2.

$$R_1 = \{ \text{Res } F(z)z^{n-1} \}_{z=a} = \frac{1}{1!} \lim_{z \rightarrow a} \frac{d}{dz} \{ (z-a)^2 F(z)z^{n-1} \}$$

Here, $a = 1$

$$R_1 = \frac{1}{1!} \lim_{z \rightarrow 1} \frac{d}{dz} \left[(z-1)^2 \frac{z^n}{(z-1)^2 (z+1)^2} \right]$$

$$= \lim_{z \rightarrow 1} \frac{d}{dz} \left[\frac{z^n}{(z+1)^2} \right]$$

$$= \lim_{z \rightarrow 1} \left[\frac{(z+1)^2 n z^{n-1} - z^n 2(z+1)(1)}{(z+1)^4} \right]$$

$$= \frac{(1+1)^2 n (1)^{n-1} - (1)^n 2(1+1)(1)}{(1+1)^4}$$

$$= \frac{4n - 4}{16}$$

$$= \frac{4(n-1)}{16}$$

$$R_1 = \frac{1}{4} (n-1)$$

Here, $a = -1$

$$\begin{aligned}
 R_2 &= \frac{1}{1!} \lim_{z \rightarrow -1} \frac{d}{dz} \left[(z+1)^2 \frac{z^n}{(z-1)^2(z+1)^2} \right] \\
 &= \lim_{z \rightarrow -1} \frac{d}{dz} \left[\frac{z^n}{(z-1)^2} \right] \\
 &= \lim_{z \rightarrow -1} \left[\frac{(z-1)^2 n z^{n-1} - z^n 2(z-1)(1)}{(z-1)^4} \right] \\
 &= \frac{(-1-1)^2 n (-1)^{n-1} - (-1)^n 2(-1-1)(1)}{(-1-1)^4} \\
 &= \frac{4n(-1)^{n-1} + (-1)^n 4}{16} \\
 &= \frac{4\{n(-1)^{n-1} + (-1)^n\}}{16} \\
 R_2 &= \frac{1}{4} \{n(-1)^{n-1} + (-1)^n\}
 \end{aligned}$$

By method of residues, $y_n = \text{Sum of Residues}$

$$= R_1 + R_2$$

$$y_n = \frac{1}{4}(n-1) + \frac{1}{4}\{n(-1)^{n-1} + (-1)^n\}$$

Example 4: Solve the difference equation $y(k+2) + y(k) = 1$, $y(0) = y(1) = 0$

Solution:

Given that, $y(k+2) + y(k) = 1$

$$y_{k+2} + y_k = 1$$

Taking Z-transform on both sides

$$Z\{y_{k+2}\} + Z\{y_k\} = Z\{1\}$$

$$z^2 \left\{ F(z) - y_0 - \frac{y_1}{z} \right\} + F(z) = \frac{z}{z-1} \quad \text{Since, } Z\{1\} = \frac{z}{z-1}$$

$$z^2 F(z) - z^2 y_0 - z y_1 + F(z) = \frac{z}{z-1}$$

Given that, $y_0 = y(0) = 0$;

$y_1 = y(1) = 0$

$$z^2 F(z) - 0 - 0 + F(z) = \frac{z}{z-1}$$

$$\{z^2 + 1\} F(z) = \frac{z}{z-1}$$

$$F(z) = \frac{z}{(z-1)(z^2+1)}$$

$$F(z) = Z\{y_k\} = \frac{z}{(z-1)(z^2+1)}$$

To find y_k using Residue Method

$$\text{Let } F(z)z^{k-1} = z^{k-1} \left\{ \frac{z}{(z-1)(z^2+1)} \right\}$$

$$F(z)z^{k-1} = \frac{z^k}{(z-1)(z^2+1)}$$

The poles of $F(z)z^{k-1}$ are given by $(z-1)(z^2+1) = 0$

$z = 1$ (Simple pole)

$$z^2 = -1$$

$z = \pm i$ (Simple pole)

$$R_1 = \{Res F(z)z^{k-1}\}_{z=a} = \lim_{z \rightarrow a} (z-a) F(z)z^{k-1}$$

Here, $a = -1$

$$R_1 = \lim_{z \rightarrow 1} (z-1) \frac{z^k}{(z-1)(z^2+1)}$$

$$= \lim_{z \rightarrow 1} \frac{z^k}{(z^2+1)} = \frac{(1)^k}{(1+1)}$$

$$R_1 = \frac{1}{2}$$

$$R_2 = \{Res F(z)z^{k-1}\}_{z=a} = \lim_{z \rightarrow a} (z-a) F(z)z^{k-1}$$

Here, $a = i$

$$R_2 = \lim_{z \rightarrow i} (z-i) \frac{z^k}{(z-1)(z^2+1)}$$

$$= \lim_{z \rightarrow i} (z-i) \frac{z^k}{(z-1)(z+i)(z-i)}$$

$$\begin{aligned}
 &= \lim_{z \rightarrow i} \frac{z^k}{(z-1)(z+i)} = \frac{i^k}{(i-1)(2i)} \\
 &= \frac{i^k}{2i(-1+i)} \times \frac{(-1-i)}{(-1-i)} \quad \text{Since } \frac{1}{i} = -i \text{ and } (a+ib)(a-ib) = a^2 + b^2 \\
 &= \frac{-i(-1-i)i^k}{2(1+1)} = \frac{i^k(i-1)}{4} \quad \text{Since } i^2 = -1
 \end{aligned}$$

$$R_2 = \frac{1}{4}(-1+i)i^k$$

To replace i by $-i$ we get, $R_3 = \frac{1}{4}(-1-i)(-i)^k$

By method of residues, $y_n = \text{Sum of Residues}$

$$= R_1 + R_2 + R_3$$

$$y_k = y(k) = \frac{1}{2} + \frac{1}{4}(-1+i)i^k + \frac{1}{4}(-1-i)(-i)^k$$

Problems for Practise

1. Using Z-transform solve $u_{n+2} - 5u_{n+1} + 6u_n = 4^n$ given $u(0) = 0$; $u(1) = 1$

$$\text{Answer: } y_n = \frac{-2}{5}(-3)^n + \frac{1}{3}(-1)^n + \frac{1}{15}(2)^n$$

2. Solve by Z-transform $y_{k+2} - 2y_{k+1} + y_k = 2^k$ with $y_0 = 2$, $y_1 = 1$

$$\text{Answer: } y_k = 1 - 2k + 2^k$$

3. Solve the difference equation $y(n) + 3y(n-1) - 4y(n-2) = 0$; $n \geq 2$ given

that $y(0) = 3$ and $y(1) = -2$

$$\text{Answer: } y(n) = (-4)^n + 2$$

Hint: Changing n to $n+2$ we get the usual form,

$$y(n+2) + 3y(n+1) - 4y(n) = 0, n \geq 0 \quad [n+2 \geq 2, n \geq 0]$$

4. Solve the difference equation $y_{n+2} - 5y_{n+1} + 6y_n = u_n$ with $y_0 = 0, y_1 = 1$,

where u_n is the unit step function.

$$\text{Hint: } Z\{\text{Unit Step Function}\} = \frac{z}{z-1} \text{ i.e., } Z\{u_n\} = \frac{z}{z-1}$$

$$\text{Answer: } \frac{1}{2} - 2(2)^n - \frac{3}{2}(3)^n$$

Application of z-transform to solve linear difference equations:

The general form of a linear difference equation of r^{th} order in the sequence y_n is

$$a_0 y_{n+r} + a_1 y_{n+r-1} + a_2 y_{n+r-2} + \dots + a_n y_n = f(n) \text{ ----->(1)}$$

When $n = 0, 1, 2 \dots$ and $a_0, a_1, \dots, a_r$ are constants

To solve the difference equations applying z-transform on both sides:

Formula:

$$Z[y_{n+1}] = Z[F(z) - y_0] \text{ where } Z[y_n] = F(z)$$

$$Z[y_{n+2}] = Z^2 \left[F(z) - y_0 - \frac{y_1}{z} \right]$$

$$Z[y_{n+3}] = Z^3 \left[F(z) - y_0 - \frac{y_1}{z} - \frac{y_2}{z^2} \right]$$

$$Z[y_{n+1}] = ZF(z) - Zy(0)$$

$$Z[y_{n+2}] = Z^2 F(z) - Z^2 y(0) - Zy(1)$$

$$Z[y_{n+3}] = Z^3 F(z) - Z^3 y(0) - Z^2 y(1) - Zy(2)$$

Problem 1: Solve $y_{n+2} + 6y_{n+1} + 9y_n = 2^n$ with $y_0 = y_1 = 0$ using z-transform

[AU 2008,2009,2010,2011]

Solution:

$$\text{Given } y_{n+2} + 6y_{n+1} + 9y_n = 2^n$$

Taking z-transform on both sides

$$Z[y_{n+2}] + 6Z[y_{n+1}] + 9Z[y_n] = Z[2^n]$$

$$Z^2 F(z) - Z^2 y(0) - Zy(1) + 6[ZF(z) - Zy(0)] + 9[F(z)] = \frac{z}{z-2} \quad \therefore Z(a)^n = \frac{z}{z-a}$$

$$\text{Given, } y_0 = y(0) = y_1 = y(1) = 0$$

$$Z^2 F(z) + 6ZF(z) + 9F(z) = \frac{Z}{z-2}$$

$$(Z^2 + 6Z + 9)F(Z) = \frac{Z}{z-2}$$

$$F(Z) = \frac{Z}{(z-2)(Z^2+6Z+9)} = \frac{Z}{(z-2)(z+3)^2}$$

$$\frac{f(z)}{z} = \frac{1}{(z-2)(z+3)^2} \text{ -----} > (1)$$

$$\text{Let } \frac{1}{(z-2)(z+3)^2} = \frac{A}{(z-2)} + \frac{B}{(z+3)} + \frac{C}{(z+3)^2}$$

$$1 = A(z+3)^2 + B(z+3)(z-2) + C(z-2)$$

$$\text{Put } z = -3 \Rightarrow 1 = A(-3+3)^2 + B(-3+3)(-3-2) + C(-3-2)$$

$$1 = -5C \Rightarrow C = -1/5$$

$$\text{Put } z = 2 \Rightarrow 1 = A(2+3)^2 + B(0) + C(0)$$

$$1 = 25A \Rightarrow A = 1/25$$

$$\text{Equating coefficients of } Z^2, \quad B = -1/25$$

(1) Becomes

$$\frac{f(z)}{z} = \frac{1}{25(z-2)} - \frac{1}{25(z+3)} - \frac{1}{5(z+3)^2}$$

$$f(z) = \frac{Z}{25(z-2)} - \frac{Z}{25(z+3)} - \frac{Z}{5(z+3)^2}$$

$$Z[y_n] = \frac{1}{25} \frac{Z}{(z-2)} - \frac{1}{25} \frac{Z}{(z+3)} - \frac{1}{5} \frac{Z}{(z+3)^2}$$

$$y_n = \frac{1}{25} z^{-1} \left[\frac{Z}{(z-2)} \right] - \frac{1}{25} z^{-1} \left[\frac{Z}{(z+3)} \right] - \frac{z^{-1}}{5} \left[\frac{Z}{(z-(-3))^2} \right]$$

$$= \frac{1}{25} 2^n - \frac{1}{25} (-3)^n - \frac{1}{5} \left(\frac{1}{-3} \right) z^{-1} \left[\frac{-3Z}{(z-(-3))^2} \right]$$

$$= \frac{1}{25} 2^n - \frac{1}{25} (-3)^n + \left(\frac{1}{15} \right) n (-3)^n \left[\begin{array}{l} \therefore z^{-1} \left[\frac{-3Z}{(z-(-3))^2} \right] = n(-3)^n \\ z^{-1} \left[\frac{az}{(z-a)^2} \right] = n(a)^n \end{array} \right]$$

$$y_n = \frac{2^n}{25} - \frac{(-3)^n}{25} + \left(\frac{n(-3)^n}{15} \right)$$

Problem 2: Solve the difference equation $y_{n+3} - 3y_{n+1} + 2y_n = 0$ given that $y(0) = 4$,

$$y(1) = 0 \text{ and } y(2) = 8$$

[AU 2007, 2011]

Solution:

$$\text{Given that } y_{n+3} - 3y_{n+1} + 2y_n = 0$$

Apply z-transform on both sides, we get

$$Z[y_{(n+3)} - 3y_{(n+1)} + 2y_{(n)}] = 0$$

$$Z[y_{(n+3)}] - 3Z[y_{(n+1)}] + 2Z[y_{(n)}] = 0$$

$$Z^3 F(z) - Z^3 y(0) - Z^2 y(1) - Zy(2) - 3Z[F(z) - y(0)] + 2F(z) = 0$$

$$\text{Given that } y(0) = 4, \quad y(1) = 0 \text{ and } y(2) = 8$$

$$Z^3 [F(z)] - 4Z^3 - Z^2(0) - 8Z - 3ZF(z) + 12Z + 2F(z) = 0$$

$$Z^3 [F(z)] - 3ZF(z) + 2F(z) = 4Z^3 + 8Z - 12Z = 4Z^3 - 4Z$$

$$(z^3 - 3z + 2) F(z) = 4Z^3 - 4Z$$

$$F(z) = \frac{4z(z^2 - 1)}{z^3 - 3z + 2}$$

$$\frac{F(z)}{z} = \frac{4z(z^2 - 1)}{z^3 - 3z + 2}$$

$$= \frac{4(z^2 - 1)}{(z-1)^2(z+2)}$$

$$= \frac{4(z+1)(z-1)}{(z-1)^2(z+2)}$$

$$\frac{F(z)}{z} = \frac{4(z+1)}{(z-1)(z+2)} \dots \dots (1)$$

$$\text{Let } \frac{4(z+1)}{(z-1)(z+2)} = \frac{A}{z-1} + \frac{B}{z+2}$$

$$4(z+1) = A(z+2) + B(z-1)$$

$$\text{Put } z = 1 \Rightarrow 4(1+1) = A(1+2) + B(0)$$

$$\Rightarrow 8 = A \cdot 3 \Rightarrow A = \frac{8}{3}$$

$$\text{Put } z = -2 \Rightarrow 4(-2+1) = A(0) + B(-2-1)$$

$$\Rightarrow -4 = -3B \Rightarrow A = \frac{4}{3}$$

$\therefore$ (1) becomes

$$\frac{F(z)}{z} = \frac{8}{3} \frac{1}{z-1} + \frac{4}{3} \frac{1}{z+2}$$

$$\Rightarrow F(z) = \frac{8}{3} \frac{z}{z-1} + \frac{4}{3} \frac{z}{z+2}$$

1	1	0	-3	-2
0	1	1	-2	0
1	1	1	-2	0
0	1	2		
1	2	0		

$(z-1)^2(z+2) \mid$

$$\Rightarrow F(z) = \frac{0}{3} \frac{z}{z-1} + \frac{4}{3} \frac{z}{z+2}$$

$$\therefore z[y(n)] = \frac{8}{3} \frac{z}{z-1} + \frac{4}{3} \frac{z}{z-(-2)}$$

$$\Rightarrow [y(n)] = \frac{8}{3} z^{-1} \left[\frac{z}{z-1} \right] + \frac{4}{3} z^{-1} \left[\frac{z}{z+2} \right]$$

$$y(n) = \frac{8}{3} (1^n) + \frac{4}{3} (-2)^n; n = 0, 1, 2, \dots$$

Problem 3: Solve using z-transform $u_{n+2} + 3u_n + 2u_n = 0$ given $u_0 = 1, u_1 = 2$ [AU 1999, 2011]

Solution: Given $u_{n+2} + 3u_{n+1} + 2u_n = 0$ and $u_0 = 1, u_1 = 2$

Applying z-transform on both sides,

$$z[u_{n+2} + 3u_{n+1} + 2u_n] = 0$$

$$\Rightarrow z[u_{n+2}] + 3z[u_{n+1}] + 2z[u_n] = 0$$

$$\Rightarrow z^2 F(z) - z^2 u_0 - zu_1 + 3zF(z) - 3zu_0 + 2F(z) = 0$$

Given $u_0 = 1, u_1 = 2$

$$\Rightarrow z^2 F(z) - z^2 u_0 - zu_1 + 3zF(z) - 3zu_0 + 2F(z) = 0$$

$$\Rightarrow z^2 F(z) - 3zF(z) + 2F(z) - z^2 - 5z = 0$$

$$\Rightarrow (z^2 + 3z + 2)F(z) = z^2 + 5z$$

$$\therefore F(z) = \frac{z(z+5)}{z^2+3z+2}$$

$$\frac{F(z)}{z} = \frac{z(z+5)}{z^2+3z+2} = \frac{(z+5)}{(z+1)(z+2)}$$

$$\text{Let } \frac{(z+5)}{(z+1)(z+2)} = \frac{A}{z+1} + \frac{B}{z+2}$$

$$z+5 = A(z+2) + B(z+1)$$

$$\text{Put } z = -1 \Rightarrow 4 = A(-2+1) + B(0) \Rightarrow$$

$$\text{Put } z = -2 \Rightarrow 3 = A(0) + B(-2+1) \Rightarrow$$

$$\frac{F(z)}{z} = \frac{4}{z+1} - \frac{3}{z+2}$$

$$A = 4$$

$$B = -3$$

$$F(z) = \frac{4z}{z+1} - \frac{3z}{z+2}$$

$$\therefore z[u_n] = 4 \frac{z}{z-(-1)} - 3 \frac{z}{z-(-2)}$$

$$u_n = 4z^{-1} \left[\frac{z}{z-1} \right] - 3z^{-1} \left[\frac{z}{z-2} \right]$$

$$u_n = 4(-1)^n - 3(-2)^n, n = 0, 1, 2, \dots$$

Problem 4: Using z-transform solve $u_{n+2} + 4u_{n+1} + 3u_n = 3^n$ with $u_0 = 0, u_1 = 1$ [AU 2010, 2011]

Solution: Given $u_{n+2} + 4u_{n+1} + 3u_n = 3^n$

Applying z-transform on both sides, $z[u_{n+2}] + 4z[u_{n+1}] + 3z[u_n] = [3^n]$

$$z^2[F(z)] - z^2 u_0 - zu_1 + 4[zF(z) - zu_0] + 3F(z) = \frac{z}{z-3}$$

$$z^2 F(z) + 4zF(z) + 3F(z) - z^2 u_0 - zu_1 + 4zu_0 = \frac{z}{z-3}$$

Given $u_0 = 0, u_1 = 1$

$$(z^2 + 4z + 3)F(z) - z^2(0) - z(1) - 4(0) = \frac{z}{z-3}$$

$$(z^2 + 4z + 3)F(z) - z = \frac{z}{z-3}$$

$$(z^2 + 4z + 3)F(z) = \frac{z}{z-3} + z = \frac{z+z^2-3z}{z-3}$$

$$(z^2 + 4z + 3)F(z) = \frac{z+z^2-3z}{z-3}$$

$$\frac{F(z)}{z} = \frac{z-2}{(z-3)(z^2+4z+3)} \rightarrow (1)$$

$$\text{Let } \frac{z-2}{(z-3)(z^2+4z+3)} = \frac{z-2}{(z-3)(z+3)(z+1)}$$

$$= \frac{A}{z+1} + \frac{B}{z+3} + \frac{C}{z-3} \rightarrow (2)$$

$$z-2 = A(z+3)(z-3) + B(z+1)(z-3) + C(z+1)(z+3)$$

$$\text{Put } z = -1 \therefore -1-2 = A(-1+3)(-1-3)$$

$$\Rightarrow -8A = -3 \Rightarrow$$

$$A = 3/8$$

$$\text{Put } z = -3 \therefore -3-2 = B(-3+1)(-3-3)$$

$$\Rightarrow -5 = 12B \Rightarrow$$

$$B = -5/12$$

$$\text{Put } z = 3$$

$$\Rightarrow 24C = 1 \Rightarrow$$

$$C = 1/24$$

$$\frac{z-2}{(z-3)(z^2+4z+3)} = \frac{3}{8} \cdot \frac{1}{z+1} + \frac{5}{12} \cdot \frac{1}{z+3} + \frac{1}{24} \cdot \frac{1}{z-3}$$

(1) becomes

$$\begin{aligned}\frac{F(x)}{z} &= \frac{3}{8} \cdot \frac{1}{z+1} - \frac{5}{12} \cdot \frac{1}{z+3} + \frac{1}{24} \cdot \frac{1}{z-3} \\ F(x) &= \frac{3}{8} \cdot \frac{z}{z+1} - \frac{5}{12} \cdot \frac{z}{z+3} + \frac{1}{24} \cdot \frac{z}{z-3} \\ z[u_n] &= \frac{3}{8} \cdot \frac{z}{z+1} - \frac{5}{12} \cdot \frac{1}{z+3} + \frac{1}{24} \cdot \frac{z}{z-3} \\ u_n &= \frac{3}{8} z^{-1} \left[\frac{z}{z+1} \right] - \frac{5}{12} z^{-1} \left[\frac{z}{z+3} \right] + \frac{1}{24} z^{-1} \left[\frac{z}{z-3} \right] \\ &= \frac{3}{8} z^{-1} \left[\frac{z}{z-(-1)} \right] - \frac{5}{12} z^{-1} \left[\frac{z}{z-(-3)} \right] + \frac{1}{24} z^{-1} \left[\frac{z}{z-3} \right] \\ \therefore u_n &= \frac{3}{8} (-1)^n - \frac{5}{12} (-3)^n + \frac{1}{24} 3^n, n = 0, 1, 2, \dots\end{aligned}$$

Problem 5: Solve $y(k+2) + y(k), y(0) = y(1) = 0$ [AU 2000, 2012]

Solution: Given $y(k+2) + y(k) = 1$

Applying z - transform

$$Z[y(k+2) + 2[y(k)]] = z[1]$$

$$z^2 = f(z) = z^2(0) - zy(1) + F(z) = \frac{z}{z-1}$$

Given $y(0) = y(1) = 0$

$$Z2 = (z) - z2(0) - z(0) + f(z) = z/z - 1 \text{ where } f(z) = z[y(K)]$$

$$(z^2 + 1)F(z) = z/z - 1$$

$$\frac{F(z)}{z} = \frac{1}{(z-1)(z^2+1)}$$

$$\frac{1}{(z-1)(z^2+1)} = \frac{A}{z-1} + \frac{Bz+c}{z^2+1}$$

$$\Rightarrow 1 = A(z^2+1) + (Bz+c)(z-1)$$

$$\text{Put } z = 1 \Rightarrow 1 A(1+1) + 0$$

$$\Rightarrow 2A = 1 \Rightarrow A = 1/2$$

$$\text{Equation coefficients of } z^2 \Rightarrow 0 = A + B \Rightarrow B = -A \Rightarrow c = -1/2$$

$$\therefore \frac{1}{(z-1)(z^2+1)} = \frac{\frac{1}{2}}{z-1} + \frac{\frac{-1}{2}z - \frac{1}{2}}{z^2+1}$$

$$\frac{F(z)}{z} = \frac{1}{2} \cdot \frac{1}{z-1} - \frac{1}{2} \cdot \frac{z}{z^2+1} - \frac{1}{2} \cdot \frac{1}{z^2+1}$$

$$\frac{F(z)}{z} = \frac{1}{2} \cdot \frac{1}{z-1} - \frac{1}{2} \cdot \frac{z}{z^2+1} - \frac{1}{2} \cdot \frac{1}{z^2+1}$$

$$\frac{F(z)}{z} = \frac{1}{2} \cdot \frac{1}{z-1} - \frac{1}{2} \cdot \frac{z^2}{z^2+1} - \frac{1}{2} \cdot \frac{1}{z^2+1}$$

$$z[y(k)] = \frac{1}{2} \cdot \frac{1}{z-1} - \frac{1}{2} \cdot \frac{z^2}{z^2+1} - \frac{1}{2} \cdot \frac{1}{z^2+1}$$

$$Y(k) = \frac{1}{2} z^{-1} \left[\frac{z}{z-1} \right] - \frac{1}{2} z^{-1} \left[\frac{z^2}{z^2+1} \right] - \frac{1}{2} z^{-1} \left[\frac{z}{z^2+1} \right]$$

$$\therefore Y(k) = \frac{1}{2} (1)^k - \frac{1}{2} \cos \frac{k\pi}{2} - \frac{1}{2} \sin \frac{k\pi}{2}$$

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Problem 6: Using z - transform solve $u_{n+2} - 5u_{n+1} + 6u_n = 4^n$ given that $u(0) = 0$; $u(1) = 1$

[AU = 2001, 2008, 09,10]

Solution: Given $u_{n+2} - 5u_{n+1} + 6u_n = 4^n$

Apply z - transform on both sides we get

$$Z[u_n + 2] = 5z[u_n + 1] + 6z[u_n] = z[4n]$$

$$Z^2 F(z) = Z^2 u_0 - zu_1 - 5[zF(z) - u_0] + 6F(z) = \frac{z}{z-4}$$

Given $u(0) = 0$; $u(1) = 1$

$$(z^2 - 5z + 6) = F(z) \cdot z^0(0) - z(1) - 5(0) = \frac{z}{z-4}$$

$$(z^2-5z+6)F(z) = \frac{z}{z-4}$$

$$(z^2-5z+6)F(z) = \frac{z}{z-4} + z$$

$$(z^2-5z+6)F(z) = \frac{z+z^2-4z}{z-4}$$

$$F(z) = \frac{z^2-3z}{(z-4)(z^2-5z+6)}$$

$$F(z) = \frac{z(z-3)}{(z-4)(z-3)(z-2)}$$

$$\frac{F(z)}{z} = \frac{1}{(z-4)(z-2)} = \frac{A}{z-4} + \frac{B}{z-2}$$

$$\Rightarrow 1 = a(z-2) + (z-4)$$

$$\text{Put } z = 2 \Rightarrow -2B \Rightarrow B = -1/2$$

$$\text{Put } z = 4 \Rightarrow 2A = 1 \Rightarrow A = 1/2$$

$$\frac{F(z)}{z} = \frac{1}{2} \cdot \frac{1}{z-4} - \frac{1}{2} \cdot \frac{1}{z-2}$$

$$\Rightarrow F(z) = \frac{1}{2(z-4)} - \frac{1}{2(z-2)}$$

$$Zu_n = \frac{1}{2(z-4)} - \frac{1}{2(z-2)}$$

$$\therefore u_n = \frac{1}{2} z^{-1} \left(\frac{z}{z-4} \right) - \frac{1}{2} z^{-1} \left(\frac{z}{z-2} \right)$$

$$\therefore u_n = \frac{1}{2} 4^n - \frac{1}{2} 2^n$$

Problem 7: Using z- transform solve $y_{n+2} + 4y_{n+1} + 3y_n = 2^n; y_0 = 0, y_1 = 1$ [AU 2010]

Hint : $z[y_{n+2}] + 4z[y_{n+1}] + 3z[y_n] = [2^n]$

Solution: Apply the formula and $y_0 = 0, y_1 = 1$

$$F(z) (z^2 + 4z + 3) - z = \frac{z}{z-2}$$

$$(z^2 + 4z + 3)F(z) = \frac{z}{z-2} + z = \frac{z+z^2-2z}{z-2} = \frac{z(z-1)}{z-2}$$

$$(z+3)(z+1)F(z) = \frac{z(z-1)}{z-2}$$

$$\frac{F(z)}{z} = \frac{(z-1)}{(z-2)(z+1)(z+3)} = \frac{(z-1)}{(z+1)(z+3)(z-2)}$$

$$\frac{F(z)}{z} = \frac{A}{z+3} + \frac{B}{z+1} + \frac{C}{z-2}$$

Solving we get A = -2/5, B = 1/3, C = 1/15

$$\therefore F(z) = -\frac{2}{5} \frac{z}{z+3} + \frac{1}{3} \frac{z}{z+1} + \frac{1}{15} \frac{z}{z-2}$$

$$Z[y_n] = \frac{-2}{5} \frac{z}{z-3} + \frac{1}{3} \frac{z}{z-1} + \frac{1}{15} \frac{z}{z-2}$$

$$y_n = \frac{-2}{5} z^{-1} \left(\frac{z}{z-3} \right) + \frac{1}{3} z^{-1} \left(\frac{z}{z-1} \right) + \frac{1}{15} z^{-1} \left(\frac{z}{z-2} \right)$$

$$\therefore y_n = \frac{-2}{5} (-3)^n + \frac{1}{3} (-1)^n + \frac{1}{15} (2)^n$$

Problem 8: Solve the difference equation $y(n) + 3y(n-1) - 4y(n-2) = 0; n \geq 2$ given that

$$y(0) = 3 \text{ and } y(1) = -2 \text{ (AU2006)}$$

Solution:

Hint: changing n to n+2 we get the usual form

$$y(n+2) + 3y(n+1) - 4y(n) = 0, n \geq 0 [n+2 \geq 2, n \geq 0]$$

Apply z-transform and given $y(0) = 3$ & $y(1) = -2$

$$(z^2 + 3z + 4)F(z) - 3z^2 + 2z + 9z = 0$$

$$F(z) = \frac{3z^2 + 7z}{z^2 + 3z + 4}$$

$$z[y(n)] = \frac{z}{z+4} + \frac{2z}{z-1}$$

$$y(n) = z^{-1} \left[\frac{z}{z-(-4)} \right] + 2z^{-1} \left[\frac{z}{z-1} \right]$$

$$y(n) = (-4)^n + 2(1^n) = (-4)^n + 2, n = 0, 1, 2, \dots$$

Problem 9: Solve by z-transform $u_{n+2} - 2u_{n+1} + u_n = 2^n$ with $u_0 = 0, u_1 = 1$ [AU 2010]

$$\text{Ans: } u_n = 2^n + 1^n - 2^n$$

Problem 10: Solve by z-transform $y_{n+2} - y_n = 2^n; y_0 = y_1 = 0$ [AU2008]

$$\text{Ans: } y_n = \frac{1}{2}(-1)^n - \frac{1}{2} + \frac{1}{3} 2^n$$

UNIVERSITY QUESTIONS

Apr. /May. 2016

Nov./Dec. 2015

Part – A

- Find the Z- Transform of $\frac{1}{n+1}$
- State the final value theorem. In Z – Transforms

Part – B

- If $U(z) = \frac{z^3+z}{(z-1)^3}$, find the value u_0, u_1 and u_2
 - Use convolution theorem to evaluate $Z^{-1}\left[\frac{z^2}{(z-3)(z-4)}\right]$
- Using the inversion integral method (Residue Theorem), find the inverse Z- Transform of $U(z) = \frac{z^2}{(z+2)(z^2+4)}$
 - Using the Z- Transform solve the difference equation $u_{n+2} + 4u_{n+1} + 3u_n = 3^n, u_0 = 0 \text{ and } u_1 = 1$

Apr. /May. 2015

Part – A

- Find $Z\left[\frac{1}{n+1}\right]$
- State the convolution theorem on Z – Transforms

Part – B

- Find $Z(r^n \sin n\theta)$, and $Z^{-1}\left[\frac{z}{z^2+4z+3}\right]$
 - Find $Z^{-1}\left[\frac{z^2}{(z-a)(z-b)}\right]$. using convolution theorem.
- Using complex residue theorem evaluate $Z^{-1}\left[\frac{9z^3}{(3z-1)^2(z-2)}\right]$
 - Solve using Z – Transform technique the difference equation $Y_{n+2} + 4Y_{n+1} + 3Y_n = 4^n$ With $y_0 = 0$ and $y_1 = 1$

Nov./Dec. 2014

Part – A

- Prove that $Z[a^n f(n)] = \bar{f}\left[\frac{z}{a}\right]$
- If $z[f(n)] = \bar{f}(z)$ then prove that $z[f(-n)] = \bar{f}\left(\frac{1}{z}\right)$

Part – B

- Find the Z – Transform of $\cos \omega t$ and $t^2 e^{-t}$
 - Solve the equation $y(n+3) - 3y(n+1) + 2y(n) = 0$ given that $y(0)=4$, $y(1)=0$ and $y(2)=8$
- State and prove final value theorem on Z – Transform
 - Using convolution theorem, find $Z^{-1}\left[\frac{8z^2}{(2z-1)(4z+1)}\right]$

Apr. /May. 2014

Part – A

- Find the Z-transform of n^2 .
- State the convolution theorem on Z – Transforms

Part – B

- Find the Z – Transform of $r^n \cos n\theta$ and $e^{-at} \cos bt$
 - Solve $u_{n+2} - 3u_{n+1} + 2u_n = 4^n, u_0 = 0$ and $u_1 = 1$
- Using convolution theorem find inverse Z – Transform of $\frac{z^2}{(z-a)(z-b)}$
 - solve: $y_{n+2} - 3y_{n+1} - 10y_n = 0, y_0 = 1$ and $y_1 = 0$.

Nov./Dec. 2013

Part – A

- Find the Z-transform of $\frac{1}{n}$.
- Find the inverse Z-transform of $\frac{z}{(z+1)^2}$.

Part – B

- (a) (i) Find the Z-transform of $\frac{1}{(n+1)(n+2)}$.
 (ii) Using Z-transform solve difference equation $Y_{n+2} + 2Y_{n+1} + Y_n = n$ given $Y_0 = 0 = Y_1$.
 (b) (i) Form the difference equation from $Y(n) = (A + Bn)2^n$.
 (ii) Using convolution theorem find $Z^{-1} \left[\frac{z^2}{(z-1)(z-3)} \right]$.

April / May 2013**Part – A**

1. Find $Z(n)$.
 2. Obtain $Z^{-1} \left[\frac{z}{(z+1)(z+2)} \right]$

Part – B

- (a) (i) Find $Z(\cos n\theta)$ and hence deduce $Z \left(\cos \frac{n\pi}{2} \right)$.
 (ii) Using Z-transform solve: $y_{n+2} - 3y_{n+1} - 10y_n = 0, y_0 = 1$ and $y_1 = 0$.
 (b) (i) State and prove the second shifting property of Z-transform.
 (ii) Using convolution theorem, find $Z^{-1} \left[\frac{z^2}{(z-a)(z-b)} \right]$.

Nov./Dec. 2012**Part – A**

1. Find the Z-transform of a^n .
 2. Solve $y_{n+1} - 2y_n = 0$, given that $y(0) = 2$.

Part – B

- (a) (i) Find the Z-transform of $\sin^2 \left(\frac{n\pi}{4} \right)$ and $\cos \left(\frac{n\pi}{2} + \frac{n\pi}{4} \right)$.
 (ii) Using convolution theorem, find the inverse Z-transform of $\frac{z^2}{(z+a)^2}$.
 (b) (i) Solve difference equation using Z-transform $y_{(n+3)} - 3y_{(n+1)} + 2y_{(n)} = 0$ given that $y_0 = 4, y_1 = 0, y_2 = 8$.
 (ii) Solve $y_{(n+2)} + 6y_{(n+1)} + 9y_{(n)} = 2^n$ given that $y_0 = y_1 = 0$.

April / May 2012**Part – A**

1. Find the Z-transform of $x(n) = \begin{cases} \frac{a^n}{n!} & \text{for } n \geq 0 \\ 0, & \text{Otherwise} \end{cases}$
 2. Solve $y_{n+1} - 2y_n = 0$, given $y_0 = 3$.

Part – B

- (a) (i) Find $Z[n(n-1)(n-2)]$.
 (ii) Using convolution theorem, find the inverse Z-transform of $\frac{8z^2}{(2z-1)(4z-1)}$.
 (b) (i) Solve $y(k+2) + y(k) = 1, y(0) = y(1) = 0$, using Z-transform.
 (ii) Solve $y_{n+2} + y_n = 2^n \cdot n$, using Z-transform.

Nov./Dec. 2011**Part – A**

1. Find the Z-transform of $\frac{1}{n!}$.
 2. Form difference equation by eliminating constants from $U_n = A 2^{n+1}$.

Part – B

- (a) (i) If $Z[f(n)] = F(z)$, find $Z[f(n-k)]$ and $Z[f(n+k)]$.
 (ii) Evaluate $Z^{-1}[(z-5)^{-3}]$ for $|z| > 5$.
 (b) (i) Solve: $u_{n+2} + 4u_{n+1} + 3u_n = 3^n, u_0 = 0$ and $u_1 = 1$.
 (ii) Form the difference equation of second order by eliminating the arbitrary constants A and B from $y_n = A(-1)^n + Bn$.

April / May 2011**Part – A**

1. Find the Z-transform of a^n .
 2. What advantage is gained when Z-transform is used to solve difference equation?

Part – B

- (a) (i) Using convolution theorem, find the inverse Z -transform of $\frac{z^2}{(z-1)(z-3)}$.
 (ii) Find the Z -transform of $a^n \cos n\theta$ and $e^{-at} \sin bt$.
- (b) (i) Solve the difference equation $y(n+3) - 3y(n+1) + 2y(n) = 0$, given that $y(0) = 4, y(1) = 0$ and $y(2) = 8$.
 (ii) Derive the difference equation from $y_n = (A + Bn)(-3)^n$.

Nov./Dec. 2010**Part – A**

1. Define the unit step sequence. Write its Z -transform.
 2. Form the difference equation by eliminating the arbitrary constants A from $y_n = A \cdot 3^n$.

Part – B

- (a) (i) Find the Z -transform of $\cos n\theta$ and $\sin n\theta$. Hence deduce the Z -transforms of $\cos(n+1)\theta$ and $a^n \sin n\theta$.
 (ii) Find the inverse Z -transform of $\frac{z(z+1)}{(z-1)^3}$ by residue method.
- (b) (i) Form the difference equation from the relation $y_n = a + b \cdot 3^n$.
 (ii) Solve: $y_{n+2} + 4y_{n+1} + 3y_n = 2^n, y_0 = 0$ and $y_1 = 1$, using Z -transform.

April / May 2010**Part – A**

1. Find the Z -transform of $\sin \frac{n\pi}{2}$.
 2. Form the difference equation generated by $y_n = a_n + b2^n$.

Part – B

- (a) (i) Solve by Z -transform $u_{n+2} - 2u_{n+1} + u_n = 2^n, u_0 = 2$ and $u_1 = 1$.
 (ii) Using convolution theorem, find the inverse Z -transform of $\left(\frac{z}{z-4}\right)^3$.
- (b) (i) Find $Z^{-1} \left[\frac{z(z^2 - z + 2)}{(z+1)(z-1)^2} \right]$ and $Z^{-1} \left[\frac{z}{(z-1)(z-2)} \right]$.
 (ii) Find $Z(na^n \sin n\theta)$.

Nov./Dec. 2009**Part – A**

1. Find the $F(z) = \frac{z^2}{(z-\frac{1}{2})(z-\frac{1}{4})(z-\frac{3}{4})}$, find $f(0)$.
 2. Find the Z -transform of $x(n) = \begin{cases} \frac{a^n}{n!} & \text{for } n \geq 0 \\ 0, & \text{otherwise} \end{cases}$

Part – B

- (a) (i) Find the inverse Z -transform of $\frac{10z}{z^2 - 3z + 2}$.
 (ii) Solve the equation $u_{n+2} + 6u_{n+1} + 9u_n = 2^n$ given $u_0 = u_1 = 0$.
- (b) (i) Using convolution theorem, find the Z^{-1} of $\frac{z^2}{(z-4)(z-3)}$.
 (ii) Find the inverse Z -transform of $\frac{z^3 - 20z}{(z-2)^3(z-4)}$.

Nov./Dec. 2011**Part – A**

1. Find the Z -transform of $\frac{a^n}{n!}$.
 2. Find $Z^{-1} \left[\frac{z}{(z-1)^2} \right]$.

Part – B

- (a) Solve the equation $y_{n+2} + 4y_{n+1} - 5y_n = 24n - 8$, given $y_0 = 3$ & $y_1 = -5$.
 (b) (i) State and prove convolution theorem on Z -transformation. Find $Z^{-1} \left[\frac{z^2}{(z-a)(z-b)} \right]$.
 (ii) If $U(z) = \frac{2z^2 + 5z + 14}{(z-1)^2}$, evaluate u_2, u_3 .